A4

Q1

a) Formulate the Dual Problem

Dual Variables:

- y_1: dual variable associated with the constraint $2x_1 + x_2 \leq 100$.
- y_2: dual variable associated with the constraint $x_1 + 3x_2 \leq 90$.

Dual Objective: Minimize the expression formed by the right-hand side of the primal constraints:

$$w = 100y_1 + 90y_2$$

Dual Constraints:

$$2y_1 + y_2 \geq 50$$
$$y_1 + 3y_2 \geq 40$$

Non-negativity: Since primal constraints are $\leq$, dual variables are:

$$y_1 \geq 0, \quad y_2 \geq 0$$

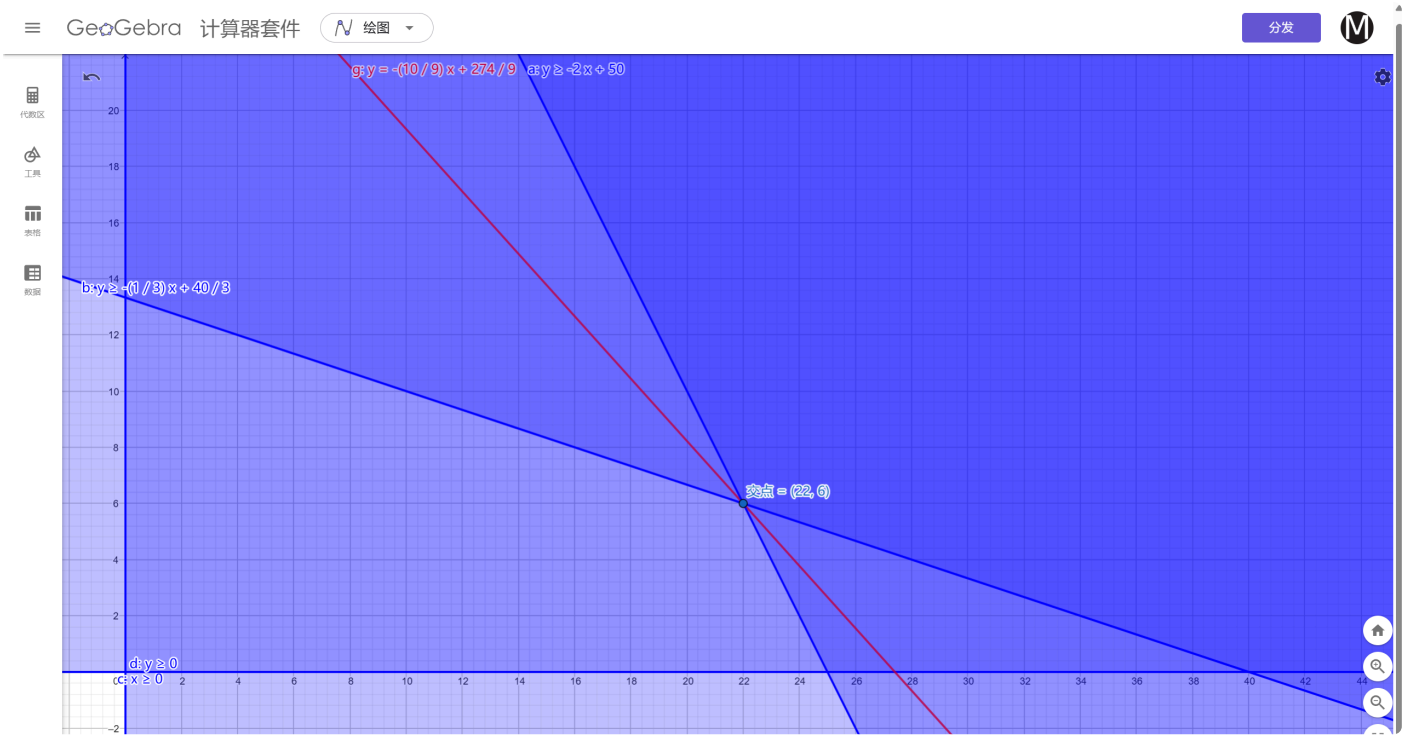
Dual Problem:

$$\text{Minimize } 100y_1 + 90y_2$$

Subject to:

$$2y_1 + y_2 \geq 50$$
$$y_1 + 3y_2 \geq 40$$
$$y_1 \geq 0, \quad y_2 \geq 0$$

b) Solve the Dual Problem Graphically



Optimal Solution: $y_1 = 22$, $y_2 = 6$, with an optimal value of 2740.

c) Verify the Weak Duality Theorem

The Weak Duality Theorem states that for any feasible primal solution $\mathbf{x}$ and dual solution $\mathbf{y}$:

$$\mathbf{c}^T\mathbf{x} \leq \mathbf{b}^T\mathbf{y}$$

Here, $50x_1 + 40x_2 \leq 100y_1 + 90y_2$.

- Primal feasible: $A\mathbf{x} \leq \mathbf{b}$, $\mathbf{x} \geq \mathbf{0}$.
- Dual feasible: $A^T\mathbf{y} \geq \mathbf{c}$, $\mathbf{y} \geq \mathbf{0}$.
- $\because A^T\mathbf{y} \geq \mathbf{c}$ and $\mathbf{x} \geq \mathbf{0}$
  - $\therefore \mathbf{x}^T A^T\mathbf{y} \geq \mathbf{x}^T\mathbf{c}$.

- $\because A\mathbf{x} \leq \mathbf{b}$ and $\mathbf{y} \geq \mathbf{0}$
  - $\therefore \mathbf{y}^T A\mathbf{x} \leq \mathbf{y}^T \mathbf{b}$.
- But $\mathbf{x}^T A^T \mathbf{y} = \mathbf{y}^T A\mathbf{x}$
  - $\therefore \mathbf{c}^T \mathbf{x} \leq \mathbf{x}^T A^T \mathbf{y} = \mathbf{y}^T A\mathbf{x} \leq \mathbf{y}^T \mathbf{b}$
  $\therefore 50x_1 + 40x_2 \leq 100y_1 + 90y_2$.

Example Verification:

- **Primal**: $x_1 = 20$, $x_2 = 20$.
  - $2 \cdot 20 + 20 = 60 \leq 100$.
  - $20 + 3 \cdot 20 = 80 \leq 90$.
  - $z = 50 \cdot 20 + 40 \cdot 20 = 1800$.
- **Dual**: $y_1 = 20$, $y_2 = 10$.
  - $2 \cdot 20 + 10 = 50 \geq 50$.
  - $20 + 3 \cdot 10 = 50 \geq 40$.
  - $w = 100 \cdot 20 + 90 \cdot 10 = 2900$.
- Check: $1800 \leq 2900$, holds.
- **Optimal Solutions**:
  - Primal max (solved similarly): $x_1 = 42$, $x_2 = 16$, $z = 2740$.
  - Dual min: $y_1 = 22$, $y_2 = 6$, $w = 2740$.
  - $2740 = 2740$, equality at optimality (strong duality), satisfying weak duality.

The theorem holds for all feasible pairs, as shown generally and via examples.

Q2

a) Formulate the Dual Problem

The dual problem is:

$$\text{Maximize} \quad 4y_1 + 9y_2$$

subject to:

$$y_1 + 2y_2 \leq 4$$
$$y_1 + 3y_2 \leq 5$$
$$y_1, y_2 \geq 0$$

b) Solve the Dual Problem Graphically



Optimal Solution: $y_1^* = 2$, $y_2^* = 1$, with an optimal value of 17.

c) Verify Primal and Dual Objective Values

- **Primal objective**:
  $4x_1 + 5x_2 = 4(3) + 5(1) = 12 + 5 = 17$
- **Dual objective** (from part b):
  $4y_1 + 9y_2 = 4(2) + 9(1) = 8 + 9 = 17$

$$\text{Primal objective value} = 17 = \text{Dual objective value}$$

This equality verifies the Strong Duality Theorem.

Q3

a) Formulating the Primal Problem

The farmer wants to minimize the total cost of purchasing fertilizers F_1 and F_2, where:

- x_1 is the quantity (in kg) of F_1 purchased, costing \$5 per kg.
- x_2 is the quantity (in kg) of F_2 purchased, costing \$4 per kg.

The **primal problem** is:

$$\text{Minimize } 5x_1 + 4x_2$$

Subject to:

$$3x_1 + x_2 \geq 30$$
$$2x_1 + 4x_2 \geq 40$$
$$x_1, x_2 \geq 0$$

b) Formulating the Dual Problem and Economic Interpretation

The **dual problem** is:

$$\text{Maximize } 30y_1 + 40y_2$$

Subject to:

$$3y_1 + 2y_2 \leq 5$$
$$y_1 + 4y_2 \leq 4$$
$$y_1 \geq 0, \quad y_2 \geq 0$$

Economic Interpretation

The dual variables y_1 and y_2 represent the **shadow prices** of nutrients A and B respectively. Since the primal minimizes cost subject to minimum nutrient requirements ($\geq$):

- y_1 is the marginal cost increase per additional kg of nutrient A required.
- y_2 is the marginal cost increase per additional kg of nutrient B required.

In economic terms, they indicate how much the total cost would rise if the farmer needed one more kg of each nutrient, reflecting their imputed value in achieving the minimum cost.

c) Solving with Python and Verifying Strong Duality

```python
from scipy.optimize import linprog

# Primal problem
c_primal = [5, 4]  # Coefficients of objective function
A_ub_primal = [[-3, -1], [-2, -4]]  # Coefficients of constraints (≤ form)
b_ub_primal = [-30, -40]  # Right-hand side of constraints

result_primal = linprog(c_primal, A_ub=A_ub_primal, b_ub=b_ub_primal)
primal_solution = result_primal.x
primal_value = result_primal.fun

print("Primal Optimal Solution (x1, x2):", primal_solution)
print("Primal Optimal Value (Minimum Cost):", primal_value)

# Dual problem
c_dual = [-30, -40]  # Minimize -(30y1 + 40y2)
A_ub_dual = [[3, 2], [1, 4]]  # Constraints (already ≤)
b_ub_dual = [5, 4]  # Right-hand side

result_dual = linprog(c_dual, A_ub=A_ub_dual, b_ub=b_ub_dual)
dual_solution = result_dual.x
dual_value = -result_dual.fun  # Negate to get maximized value

print("Dual Optimal Solution (y1, y2):", dual_solution)
print("Dual Optimal Value (Maximum Value):", dual_value)

# Verify Strong Duality
difference = primal_value - dual_value
print("Difference (Primal - Dual):", difference)
if abs(difference) < 1e-10:
```

```
    print("Strong Duality Theorem is satisfied (difference ≈ 0)")
else:
    print("Strong Duality Theorem may not hold (check computations)")
```

Results :

```
PS D:\MG\! CUHKSZ\~! 大二 下\MAT3007\Assignments\4> & D:/MG/Python/python.exe "d:/MG/! CUHKSZ/~! 大二
下/MAT3007/Assignments/4/Q3.py"
Primal Optimal Solution (x1, x2): [8. 6.]
Primal Optimal Value (Minimum Cost): 64.0
Dual Optimal Solution (y1, y2): [1.2 0.7]
Dual Optimal Value (Maximum Value): 64.0
Difference (Primal - Dual): 0.0
Strong Duality Theorem is satisfied (difference ≈ 0)
```

Q4

a) Formulating the Primal Problem

Define x_{ij} as the number of units shipped from warehouse i (where $i = 1$ for W_1, $i = 2$ for W_2) to store j (where $j = 1$ for S_1, $j = 2$ for S_2, $j = 3$ for S_3).
$\because$ Can't ship negative quantities
$\therefore x_{ij} \geq 0$

The **primal problem** is:

$$\text{Minimize } Z = 6x_{11} + 8x_{12} + 10x_{13} + 7x_{21} + 5x_{22} + 9x_{23}$$

Subject to:

$$\begin{aligned}
x_{11} + x_{12} + x_{13} &\leq 500 \quad (W_1 \text{ supply}) \\
x_{21} + x_{22} + x_{23} &\leq 400 \quad (W_2 \text{ supply}) \\
x_{11} + x_{21} &\geq 300 \quad (S_1 \text{ demand}) \\
x_{12} + x_{22} &\geq 400 \quad (S_2 \text{ demand}) \\
x_{13} + x_{23} &\geq 200 \quad (S_3 \text{ demand}) \\
x_{ij} &\geq 0 \text{ for all } i, j
\end{aligned}$$

b) Formulating the Dual Problem and Interpreting Dual Variables

Assign dual variables to each primal constraint:

- u_1, u_2: for supply constraints of W_1 and W_2
- v_1, v_2, v_3: for demand constraints of S_1, S_2, S_3

The **dual problem** is:

$$\text{Maximize } 500u_1 + 400u_2 + 300v_1 + 400v_2 + 200v_3$$

Subject to:

$$\begin{aligned}
u_1 + v_1 &\leq 6 \\
u_1 + v_2 &\leq 8 \\
u_1 + v_3 &\leq 10 \\
u_2 + v_1 &\leq 7 \\
u_2 + v_2 &\leq 5 \\
u_2 + v_3 &\leq 9 \\
u_1, u_2 \leq 0, &\quad v_1, v_2, v_3 \geq 0
\end{aligned}$$

Interpretation of Dual Variables

- u_i: Shadow price of the supply constraint for warehouse i. It represents the change in total cost per unit increase in supply at W_i. A positive u_i suggests that increasing supply reduces cost (e.g., access to cheaper routes), while a negative u_i indicates increased cost.
- v_j: Shadow price of the demand constraint for store j. It represents the change in total cost per unit increase in demand at S_j. A positive v_j indicates the marginal cost of meeting additional demand, while a negative v_j suggests cost savings if demand decreases.

In transportation problems, $u_i + v_j = c_{ij}$ holds for routes used ($x_{ij} > 0$), reflecting the cost allocation between supply and demand points.

c) Solving with Python and Verifying Strong Duality

```
from pulp import *

# Solve the Primal Problem

# Define the primal problem
primal = LpProblem("Transportation_Primal", LpMinimize)

# Define decision variables with non-negativity
x11 = LpVariable("x11", lowBound=0)
x12 = LpVariable("x12", lowBound=0)
x13 = LpVariable("x13", lowBound=0)
x21 = LpVariable("x21", lowBound=0)
x22 = LpVariable("x22", lowBound=0)
x23 = LpVariable("x23", lowBound=0)

# Set the objective function
primal += 6*x11 + 8*x12 + 10*x13 + 7*x21 + 5*x22 + 9*x23, "Total_Cost"

# Add supply constraints
primal += x11 + x12 + x13 <= 500, "W1_Supply"
primal += x21 + x22 + x23 <= 400, "W2_Supply"

# Add demand constraints
primal += x11 + x21 >= 300, "S1_Demand"
primal += x12 + x22 >= 400, "S2_Demand"
primal += x13 + x23 >= 200, "S3_Demand"

# Solve the primal problem
primal.solve()

# Print the results
print("Primal Problem Results:")
print("Status:", LpStatus[primal.status])
print("Optimal Solution:")
for v in primal.variables():
    print(f"{v.name} = {v.varValue}")
print("Total Cost (Primal Objective) =", value(primal.objective))

# Solve the Dual Problem

# Define the dual problem
dual = LpProblem("Transportation_Dual", LpMaximize)

# Define dual variables with appropriate bounds
u1 = LpVariable("u1", upBound=0)  # u1 <= 0
u2 = LpVariable("u2", upBound=0)  # u2 <= 0
v1 = LpVariable("v1", lowBound=0)  # v1 >= 0
v2 = LpVariable("v2", lowBound=0)  # v2 >= 0
v3 = LpVariable("v3", lowBound=0)  # v3 >= 0

# Set the objective function
dual += 500*u1 + 400*u2 + 300*v1 + 400*v2 + 200*v3, "Dual_Objective"

# Add constraints
dual += u1 + v1 <= 6, "Constraint_11"
dual += u1 + v2 <= 8, "Constraint_12"
dual += u1 + v3 <= 10, "Constraint_13"
dual += u2 + v1 <= 7, "Constraint_21"
dual += u2 + v2 <= 5, "Constraint_22"
dual += u2 + v3 <= 9, "Constraint_23"

# Solve the dual problem
dual.solve()

# Print the results
print("\nDual Problem Results:")
print("Status:", LpStatus[dual.status])
print("Optimal Solution:")
for v in dual.variables():
    print(f"{v.name} = {v.varValue}")
print("Dual Objective =", value(dual.objective))
```

Results:

```
PS D:\MG\! CUHKSZ\~! 大二 下\MAT3007\Assignments\4> & D:/MG/Python/python.exe "d:/MG/! CUHKSZ/~! 大二
下/MAT3007/Assignments/4/Q4.py"
Welcome to the CBC MILP Solver
Version: 2.10.3
Build Date: Dec 15 2019

command line - D:\MG\Python\Lib\site-packages\pulp\solverdir\cbc\win\64\cbc.exe
C:\Users\19927\AppData\Local\Temp\57f0f34b07ab4f26b6e604a404f8f8f4-pulp.mps -timeMode elapsed -branch -printingOptions all -
```

```
solution C:\Users\19927\AppData\Local\Temp\57f0f34b07ab4f26b6e604a404f8f8f4-pulp.sol (default strategy 1)
At line 2 NAME          MODEL
At line 3 ROWS
At line 10 COLUMNS
At line 29 RHS
At line 35 BOUNDS
At line 36 ENDATA
Problem MODEL has 5 rows, 6 columns and 12 elements
Coin0008I MODEL read with 0 errors
Option for timeMode changed from cpu to elapsed
Presolve 5 (0) rows, 6 (0) columns and 12 (0) elements
0  Obj 0 Primal inf 900 (3)
4  Obj 5800
Optimal - objective value 5800
Optimal objective 5800 - 4 iterations time 0.002
Option for printingOptions changed from normal to all
Total time (CPU seconds):       0.00   (Wallclock seconds):       0.00

Primal Problem Results:
Status: Optimal
Optimal Solution:
x11 = 300.0
x12 = 0.0
x13 = 200.0
x21 = 0.0
x22 = 400.0
x23 = 0.0
Total Cost (Primal Objective) = 5800.0
Welcome to the CBC MILP Solver
Version: 2.10.3
Build Date: Dec 15 2019

command line - D:\MG\Python\Lib\site-packages\pulp\solverdir\cbc\win\64\cbc.exe
C:\Users\19927\AppData\Local\Temp\ab728159b86f42a28dd36520040d9d2d-pulp.mps -max -timeMode elapsed -branch -printingOptions all -
solution C:\Users\19927\AppData\Local\Temp\ab728159b86f42a28dd36520040d9d2d-pulp.sol (default strategy 1)
At line 2 NAME          MODEL
At line 3 ROWS
At line 11 COLUMNS
At line 29 RHS
At line 36 BOUNDS
At line 41 ENDATA
Problem MODEL has 6 rows, 5 columns and 12 elements
Coin0008I MODEL read with 0 errors
Option for timeMode changed from cpu to elapsed
Presolve 6 (0) rows, 5 (0) columns and 12 (0) elements
0  Obj -0 Dual inf 900 (3)
0  Obj -0 Dual inf 900 (3)
4  Obj 5800
Optimal - objective value 5800
Optimal objective 5800 - 4 iterations time 0.002
Option for printingOptions changed from normal to all
Total time (CPU seconds):       0.00   (Wallclock seconds):       0.00


Dual Problem Results:
Status: Optimal
Optimal Solution:
u1 = 0.0
u2 = -1.0
v1 = 6.0
v2 = 6.0
v3 = 10.0
Dual Objective = 5800.0
```

Q5

a) Write the Dual Problem

The **dual problem** is:

$$\text{maximize} \quad 0$$

Subject to:

$$Ax = b$$
$$x \geq 0$$

b) Use Weak and Strong Duality Theorems

(1) If $b^T y < 0$ for a feasible y, then there cannot exist x such that $Ax = b$ and $x \geq 0$.

A feasible y satisfies $A^T y \geq 0$ with $b^T y < 0$.
Suppose there exists $x \geq 0$ such that $Ax = b$, making the dual feasible.

Compute:

$$x^T(A^Ty) = (Ax)^Ty = b^Ty.$$

$\because x \geq 0$ and $A^Ty \geq 0$, each term $x_i(A^Ty)_i \geq 0$:

$$\therefore \quad x^TA^Ty = b^Ty \geq 0.$$

But we're given $b^Ty < 0$, leading to a contradiction.
Thus, no such x exists.

This aligns with weak duality: if the dual is feasible ($0^Tx = 0$), then $0 \leq b^Ty$ for all feasible y, but $b^Ty < 0$ violates this, suggesting the dual is infeasible.

(2) If $b^Ty \geq 0$ for all feasible y, then there must exist x such that $Ax = b$ and $x \geq 0$.

Assume $b^Ty \geq 0$ for all y such that $A^Ty \geq 0$.
In the primal: $\because y = 0$ is feasible ($A^T0 = 0 \geq 0$, $b^T0 = 0$), and $b^Ty \geq 0$ for all feasible y
$\therefore$ The infimum is at least 0, and is achieved at $y = 0$ with $b^T0 = 0$.
The feasible set $\{y \mid A^Ty \geq 0\}$ is a polyhedral cone, and the linear objective is bounded below by 0, so the primal has an optimal value of 0.

By strong duality, if the primal has a finite optimal value (here, 0) and is feasible, and if the dual is feasible, then the dual's optimal value equals the primal's. The dual's objective is $0^Tx = 0$, so if feasible, its value is 0, matching the primal's 0. We need to show the dual is feasible.

If the dual were infeasible (no $x \geq 0$ such that $Ax = b$), consider the primal's behavior. Since the primal is feasible and achieves 0, strong duality implies the dual cannot be infeasible when the primal is bounded and feasible. Thus, there must exist $x \geq 0$ such that $Ax = b$.

c) Prove Farkas' Lemma

Farkas' Lemma states exactly one of these is true:

1. There exists $x \in \mathbb{R}^n$ such that $Ax = b$ and $x \geq 0$.
2. There exists $y \in \mathbb{R}^m$ such that $A^Ty \geq 0$ and $b^Ty < 0$.

- **Mutual Exclusivity**: From `b)` `(1)`, if `2` holds ($A^Ty \geq 0$, $b^Ty < 0$), then `1` cannot hold—no $x \geq 0$ satisfies $Ax = b$. So, `1` and `2` cannot both be true.
- **Exhaustiveness**: From `b)` `(2)`, if $b^Ty \geq 0$ for all y with $A^Ty \geq 0$, then `1` holds (there exists $x \geq 0$ such that $Ax = b$). Conversely, if `1` does not hold (no such x), then it's not true that $b^Ty \geq 0$ for all feasible y, so there exists y with $A^Ty \geq 0$ and $b^Ty < 0$, i.e., `2` holds.

Thus, either `1` is true and `2` is false, or `2` is true and `1` is false, proving Farkas' Lemma.